

1. Missing Database Driver

To cause error: Remove MySQL connector from pom.xml

Error message:

```
java.lang.ClassNotFoundException: com.mysql.cj.jdbc.Driver
    at java.base/jdk.internal.loader.BuiltinClassLoader.loadClass(BuiltinClassLoader.java:641)
    at
    java.base/jdk.internal.loader.ClassLoaders$AppClassLoader.loadClass(ClassLoaders.java:188)
    at java.base/java.lang.Class.forName0(Native Method)
    at java.base/java.lang.Class.forName(Class.java:501)
    at your.package.DatabaseConnection.initializeDriver(DatabaseConnection.java:25)
```

2. Database Unreachable

To cause error: change the database url to an incorrect port

Error message:

```
com.mysql.cj.jdbc.exceptions.CommunicationsException: Communications link failure
The last packet sent successfully to the server was 0 milliseconds ago. The driver has not
received any packets from the server.
    at
    com.mysql.cj.jdbc.exceptions.SQLError.createCommunicationsException(SQLError.java:174)
    at
    com.mysql.cj.jdbc.exceptions.SQLExceptionsMapping.translateException(SQLExceptionsMapping.java:64)
Caused by: java.net.ConnectException: Connection refused (Connection refused)
    at java.base/java.net.PlainSocketImpl.socketConnect(Native Method)
    at java.base/java.net.AbstractPlainSocketImpl.doConnect(AbstractPlainSocketImpl.java:406)
```

3. Database Not Authorized

To cause error: use incorrect database credentials such as username or password

Error message:

```
java.sql.SQLException: Access denied for user 'invalid_user'@'localhost' (using password: YES)
    at com.mysql.cj.jdbc.exceptions.SQLError.createSQLException(SQLError.java:129)
    at
com.mysql.cj.jdbc.exceptions.SQLExceptionsMapping.translateException(SQLExceptionsMapping.java:122)
    at com.mysql.cj.jdbc.ConnectionImpl.createNewIO(ConnectionImpl.java:825)
    at com.mysql.cj.jdbc.ConnectionImpl.<init>(ConnectionImpl.java:445)
```

4. Syntax error in SQL Statement

To cause error: add a typo or other invalid syntax to your sql code

Error message:

```
java.sql.SQLSyntaxErrorException: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'SELECCT * FROM users' at line 1
    at com.mysql.cj.jdbc.exceptions.SQLError.createSQLException(SQLError.java:120)
    at
com.mysql.cj.jdbc.exceptions.SQLExceptionsMapping.translateException(SQLExceptionsMapping.java:122)
    at com.mysql.cj.jdbc.StatementImpl.executeQuery(StatementImpl.java:1187)
    at your.package.UserDAO.getAllUsers(UserDAO.java:45)
```

5. Error Extracting Data from Result Set (bad column name)

To cause error: reference a column in your database that does not exist

Error message:

```
java.sql.SQLException: Column 'username' not found.
    at com.mysql.cj.jdbc.exceptions.SQLError.createSQLException(SQLError.java:129)
    at
com.mysql.cj.jdbc.exceptions.SQLExceptionsMapping.translateException(SQLExceptionsMapping.java:122)
    at com.mysql.cj.jdbc.result.ResultSetImpl.findColumn(ResultSetImpl.java:550)
    at com.mysql.cj.jdbc.result.ResultSetImpl.getString(ResultSetImpl.java:887)
```

at your.package.UserDAO.mapResultSetToUser(UserDAO.java:78)

6. Error Extracting Data from Result Set (wrong data type)

To cause error: call getInt on a varchar data type in your database

Error message:

```
java.sql.SQLException: Cannot convert value 'John Doe' from column 2 to INT.  
    at com.mysql.cj.jdbc.exceptions.SQLError.createSQLException(SQLError.java:129)  
    at  
com.mysql.cj.jdbc.exceptions.SQLExceptionsMapping.translateException(SQLExceptionsMapping.java:122)  
    at com.mysql.cj.jdbc.result.ResultSetImpl.getInt(ResultSetImpl.java:795)  
    at your.package.UserDAO.mapResultSetToUser(UserDAO.java:82)
```